

Problem Statement

Current State:

Consumers are inundated with irrelevant promotions and often miss timely, valuable offers that match their immediate context and preferences. Retailers and service providers struggle to drive foot traffic and conversions with generic, untargeted campaigns.

Impact & Affected Stakeholders:

- Urban professionals aged 18-45
- Frequent shoppers using smartphones for payments and navigation
- Retailers and service providers seeking higher conversion rates

Why Now:

The proliferation of mobile devices and digital payments has increased consumer expectations for personalized, real-time engagement. Merchants face growing competition and need more effective ways to reach and convert customers.

Quantitative Scope:

Low offer redemption rates (<5%) for generic campaigns; potential to increase redemption rates to 15-20% with personalized, context-aware offers.

Proposed Solution

Approach:

Develop a mobile app that integrates geofencing, behavioral analytics, and transaction data to deliver targeted, real-time offers to users based on their location, activity, and spending patterns. Offers are delivered via push notifications or in-app, with seamless redemption at partner merchants.

Key Decisions:

- Utilize real-time location tracking with user consent for precise offer delivery
- Integrate with payment providers to analyze transaction history securely
- Allow users to control notification preferences (always-on vs. app-open only)
- Partner with merchants for exclusive, redeemable offers

Trade-offs:

- Balancing user privacy with the need for continuous location tracking
- Potential battery drain from real-time geofencing
- Limiting initial merchant partnerships to ensure offer quality and relevance

Target Users & Personas

Urban Professional _(Full-time employee, city dweller)_

Goals:

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- Save money on everyday purchases
- Discover relevant deals nearby
- Minimize time spent searching for offers

Pain Points:

- Receives too many irrelevant promotions
- Misses out on timely deals
- Does not want to manually search for offers

Behavior Patterns:

- Uses smartphone for payments and navigation
- Shops and dines out frequently
- Open to trying new merchants if incentivized

Frequent Shopper _(Retail and dining enthusiast)_

Goals:

- Maximize savings
- Get personalized recommendations
- Redeem offers easily

Pain Points:

- Overwhelmed by generic deal apps
- Frustrated by expired or irrelevant offers
- Wants seamless redemption experience

Behavior Patterns:

- Checks for deals before making purchases
- Shares offers with friends
- Prefers instant, mobile-based redemptions

Merchant Marketer _(Retail or restaurant marketing manager)_

Goals:

- Increase foot traffic
- Improve offer redemption rates
- Access actionable campaign analytics

Pain Points:

- Low ROI from generic campaigns
- Difficulty targeting the right audience
- Limited insights into customer behavior

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Behavior Patterns:

- Runs multiple campaigns across channels
- Seeks platforms with measurable results
- Willing to pay for premium placement

User Stories & Use Cases

US1 — Priority: High

As a **User**, Receive a relevant dining offer when near a restaurant during lunch hours, Saves money and discovers new places

US2 — Priority: High

As a **User**, Control whether the app tracks location always or only when open, Maintains privacy and battery life

US3 — Priority: High

As a **User**, Redeem an offer instantly at checkout via mobile, Seamless and fast experience

US4 — Priority: Medium

As a **User**, View offer history and redemption status, Tracks savings and avoids duplicate redemptions

US5 — Priority: High

As a **User**, Receive only offers relevant to my spending habits, Avoids irrelevant notifications

US6 — Priority: Medium

As a **User**, Not receive offers when outside supported cities, Prevents irrelevant notifications

US7 — Priority: Medium

As a **Merchant**, Access analytics on offer performance and user engagement, Optimizes future campaigns

US8 — Priority: Low

As a **User**, Report an expired or invalid offer, Improves offer quality

US9 — Priority: Medium

As a **User**, Receive a notification only once per offer per day, Reduces notification fatigue

US10 — Priority: High

As a **User**, Opt out of data sharing at any time, Maintains control over privacy

Functional Requirements

Must Have:

- **FR1:** The app must deliver location-based offers to users in real time based on their current location and time of day.
- **FR2:** The app must allow users to control location tracking settings (always-on, app-open only, or off).
- **FR3:** The app must analyze user transaction history to personalize offer recommendations.
- **FR4:** Users must be able to redeem offers instantly via a mobile interface at participating merchants.

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- **FR5:** The app must support push notifications for relevant offers.
- **FR6:** Users must be able to view their offer history and redemption status.
- **FR7:** The app must allow users to opt out of data sharing and delete their account/data.

Should Have:

- **FR8:** The app should allow users to report expired or invalid offers.
- **FR9:** The app should limit offer notifications to one per offer per day per user.
- **FR10:** Merchants should have access to a dashboard with analytics on offer performance and user engagement.

Nice to Have:

- **FR11:** The app could support social sharing of offers with friends.
- **FR12:** The app could provide personalized recommendations based on user reviews and ratings.
- **FR13:** The app could integrate with wearable devices for proximity-based notifications.

Non-Functional Requirements

Performance:

Offer notifications must be delivered within 2 seconds of a qualifying event (e.g., entering a geofenced area). App screens must load within 1 second on devices from the last 3 years.

Security:

All user data must be encrypted in transit and at rest. The app must comply with GDPR and CCPA for data privacy and user consent.

Scalability:

The platform must support at least 1 million concurrent users and 10,000 simultaneous merchant campaigns without degradation.

Accessibility:

The app must meet WCAG 2.1 AA standards, including support for screen readers, high-contrast modes, and adjustable font sizes.

Reliability:

The system must maintain 99.9% uptime, with automated failover and recovery for critical services.

Scope & Constraints

In Scope:

- Mobile app for iOS and Android
- Real-time location-based offer delivery
- User transaction data integration
- Push notifications
- Merchant dashboard for analytics

Out of Scope:

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- Web-based user interface for consumers
- Physical coupon printing
- Integration with non-partner merchants

Constraints:

- User consent required for location and transaction data access
- Initial launch limited to select urban markets
- Merchant onboarding subject to quality review

Success Metrics & KPIs

Offer Redemption Rate

- Baseline: 5%
- Target: 15% within 6 months
- Measurement: Percentage of delivered offers redeemed by users
- Timeline: Monthly

Monthly Active Users (MAU)

- Baseline: 0 (pre-launch)
- Target: 100,000 within 12 months
- Measurement: Unique users engaging with the app per month
- Timeline: Monthly

Merchant Conversion Rate

- Baseline: N/A
- Target: 10% of merchants running repeat campaigns within 6 months
- Measurement: Percentage of merchants renewing or expanding campaigns
- Timeline: Quarterly

User Churn Rate

- Baseline: N/A
- Target: <10% monthly churn
- Measurement: Percentage of users uninstalling or becoming inactive
- Timeline: Monthly

Average Offer Notification Delivery Time

- Baseline: N/A
- Target: <2 seconds
- Measurement: System logs of notification delivery latency
- Timeline: Weekly

Dependencies & Risks

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Dependencies:

- Integration with payment providers for transaction data
- Partnership agreements with merchants
- Mobile OS location services APIs
- Push notification services (APNs, FCM)

Risks:

- **User privacy concerns leading to low opt-in rates for location and data sharing**

- Likelihood: Medium | Impact: High

- Mitigation: Transparent consent flows, clear privacy policy, and granular controls

- **Battery drain from continuous location tracking**

- Likelihood: Medium | Impact: Medium

- Mitigation: Optimize geofencing logic, allow user control over tracking frequency

- **Merchant offer quality inconsistency**

- Likelihood: Medium | Impact: Medium

- Mitigation: Merchant onboarding review, user reporting of invalid offers

- **Regulatory changes affecting data usage**

- Likelihood: Low | Impact: High

- Mitigation: Monitor legal landscape, design for compliance

- **Scalability issues during rapid user growth**

- Likelihood: Low | Impact: High

- Mitigation: Cloud-based infrastructure with auto-scaling

Acceptance Criteria

AC1: Location-based offer delivery

- GIVEN A user with location tracking enabled enters a geofenced area during merchant offer hours
- WHEN The user is detected within the geofence
- THEN:

- A relevant offer notification is delivered to the user's device within 2 seconds

- The offer matches the user's preferences and transaction history

AC2: User controls location tracking

- GIVEN A user is on the app settings page
- WHEN They select 'Only track location when app is open'
- THEN:

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- The app only accesses location data when the app is in the foreground

AC3: Offer redemption

- GIVEN A user has received an offer notification
 - WHEN They tap the offer and present it at a participating merchant
 - THEN:
- The offer is marked as redeemed in the app
 - The merchant receives confirmation of redemption

AC4: User views offer history

- GIVEN A user is logged in
 - WHEN They navigate to the offer history screen
 - THEN:
- All redeemed and active offers are displayed with status and date

AC5: User opts out of data sharing

- GIVEN A user is on the privacy settings page
 - WHEN They disable data sharing
 - THEN:
- The app stops collecting transaction and location data
 - Personalized offers are disabled

AC6: Reporting invalid offers

- GIVEN A user attempts to redeem an expired or invalid offer
 - WHEN They tap 'Report Issue'
 - THEN:
- The offer is flagged for review
 - The user receives confirmation of report submission

AC7: Notification frequency control

- GIVEN A user has already received a notification for a specific offer today
 - WHEN They re-enter the same geofenced area
 - THEN:
- No additional notification for that offer is sent on the same day

AC8: Accessibility compliance

- GIVEN A user with a screen reader enabled opens the app
 - WHEN They navigate through the main screens
 - THEN:
- All interactive elements are accessible and labeled appropriately

AC9: Merchant analytics dashboard

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- GIVEN A merchant logs into the dashboard
- WHEN They view campaign analytics
- THEN:
 - Metrics such as impressions, redemptions, and user engagement are displayed accurately

AC10: Data deletion

- GIVEN A user requests account deletion
- WHEN The request is confirmed
- THEN:
 - All user data is permanently deleted from the system within 30 days

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